TASK 6: ADVERSARY EMULATION WITH MITRE CALDERA

Analyst: SOC Analyst
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Status: Complete

1. OBJECTIVE

Simulate adversary TTPs using MITRE Caldera and validate SOC detection capabilities via Elastic Security.

2. LAB SETUP

Component	Detail
Caldera Server	192.168.56.130:8888
Agent	Sandcat (jbpors)
Target Host	Window-Endpoint
SIEM	Elastic Stack (Kibana)
Adversary Profile	Discovery

3. EMULATION EXECUTION

- Deployed Sandcat agent on Windows Endpoint
- Created operation with Discovery adversary profile
- Executed 7 discovery techniques
- Monitored detection in Elastic Security

4. TECHNIQUES SIMULATED

Tactic: Discovery (TA0007)

ABILITY	STATUS
IDENTIFY FIREWALLS	Success
DISCOVER DOMAIN CONTROLLER	Failed (No DC in lab)
VIEW ADMIN SHARES	Success
IDENTIFY ACTIVE USER	Success
IDENTIFY LOCAL USERS	Success
DISCOVER ANTIVIRUS PROGRAMS	Success
PERMISSION GROUPS DISCOVERY	Success

Result: 6/7 Successful, 1 Failed (Expected)

5. DETECTION VALIDATION

Query: agent.name: "Window-Endpoint" AND event.code: "4688"

Metric	Value
Event ID	4688 (Process Creation)
Events Captured	11
Detection Status	SUCCESS

All emulation activities logged and visible in Kibana Discover during operation window (18:24–18:43).

6. MITRE ATT&CK MAPPING

Tactic	Techniques
TA0007 — Discovery	T1087, T1069, T1518, T1016, T1057

7. KEY FINDINGS

- Caldera successfully emulated discovery-phase adversary behavior
- Elastic Security detected all process creation events
- DC discovery failed due to lab environment limitations (expected)
- 11 Event 4688 records provide complete audit trail

8. SCREENSHOTS

File	Description
agent_alive.png	Sandcat agent connected to Caldera
operation_running.png	Operation steps executing
operation_completed.png	Final status with Success/Failed
elastic_detection.png	Kibana showing 4688 events